

TABLE VI
Drying Methods*

Circulating air/convection	Convection	- Large space required - Long drying times - Only approx. 5 per cent of energy used
IR radiation	Radiation/absorption 0.76–1.000 μm	- Short drying times - High heat transfer - Direct heating of the coating layers
UV radiation	Radiation/absorption 100–400 nm	- Fastest curing - Almost no heating of the substrate - Shadow effect
Electron beam radiation	Radiation/absorption	Almost never used in industry
Microwave drying	Radiation/absorption e.g.: 2.456 MHz = 12.24 cm	- Direct heating of the solvent/dispersing agent - Well-suited for aqueous coatings - Conductors may act as antenna and overheat (400°C)

*Based on Rippert Anlagentechnik, Interfinish 2000

Removing residues which may affect adhesion and insulating properties, secure and flawless processing as well as complete curing are essential factors if optimal insulation properties are to be achieved, and allowing a high-performance conformal coating. Prior to selecting an application method that constitutes the optimal procedure for the user, it must be

clarified which frame conditions have to be fulfilled. In this context, the necessary or potential production capacities are of importance, along with the questions whether a selective coating is required and to what extent the process will be automated, and if contract coating is intended.

Fig. 4 shows a flow-chart of the conformal coating process. The data of

this flow-chart may change depending on any specific requirement.

Drying

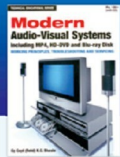
A conformal coating can satisfy the set requirements only if it is thoroughly dried and/or cured. For this reason, it is essential to check the drying and curing parameters of each type of assembly to be coated (refer Table VI). With room-temperature drying, a check of the final physical and electrical film properties would make sense after 96 hours at the earliest, in accordance with DIN 46449. A functional test (burn-in test) of the coated assembly is certainly possible beforehand if the coating of the assembly is manually or mechanically resistant. In the case of aqueous coatings, this is possible only to a limited extent, as moisture and thus electrically conductive areas can be maintained for some time. ●

This article is an extract from the book titled 'Conformal Coatings for Electronics Applications' by the author. The sources of information are mentioned in the book

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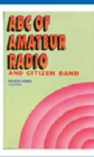
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